Cross-Generator Generalization in AI-Generated Image Detection Using ResNet50

DATS 6303 – Deep Learning and Fundamentals of AI.

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Problem and Motivation

Generative models such as Stable Diffusion and Midjourney now produce photorealistic images that can be misused for misinformation and fraud. Neural networks can learn to separate real from generated images, but detectors trained on one generator often fail on generators unseen during training. Because new generators appear continually, this weakness limits practical use. We will investigate whether a ResNet50 detector trained on one generator can identify images from unseen generators.

Dataset

We will use GenImage (Zhu et al., 2023), a public benchmark of approximately 2.68 million images: 1.33 million real ImageNet images and 1.35 million images from eight generators. We plan to train on the Stable Diffusion v1.4 subset (roughly 300,000 images) and test on Midjourney, ADM, GLIDE, VQDM, Wukong, and BigGAN. This scale is sufficient to fine-tune a deep network. Download feasibility and exact subset sizes are being verified.

Model

We will fine-tune an ImageNet-pretrained ResNet50 (He et al., 2016) end to end, replacing the 1,000 class output layer with a single logit trained with binary cross-entropy. All layers will be updated. To reduce shortcut learning from compression and size differences, both classes will share a common JPEG re-encoding and cropping protocol.

Framework

We will use PyTorch and torchvision, which provide pretrained weights, mixed-precision GPU training, and transparent training loops we can modify and explain.

GPU Requirement

Fully fine-tuning a 25-million-parameter ResNet50 model on approximately 300,000 images per epoch is computationally intensive and impractical on CPU. We have already configured an AWS EC2 g5.xlarge instance equipped with an NVIDIA A10G GPU (24 GB VRAM) for model training. We will use GPU acceleration and report the hardware specifications, training throughput, and a measured CPU comparison.

Experiments and Metrics

We will evaluate in-distribution performance on held-out Stable Diffusion v1.4 images, cross-generator performance on unseen generators, and robustness to JPEG compression and resizing. We will report accuracy, precision, recall, F1-score, and ROC-AUC. The decision threshold will be fixed on the training generator's validation data. The generalization gap is the drop in ROC-AUC on each unseen generator.

References

Zhu, M., et al. (2023). GenImage: A Million-Scale Benchmark for Detecting AI-Generated Image. NeurIPS.

He, K., et al. (2016). Deep Residual Learning for Image Recognition. CVPR.

Wang, S.-Y., et al. (2020). CNN-Generated Images Are Surprisingly Easy to Spot… for Now. CVPR.

GenImage repository: https://github.com/weili-thu/genimage

PyTorch documentation: https://pytorch.org/docs

Timeline

Week 1: Verify data and GPU access; build data and training pipelines.

Week 2: Baseline training.

Week 3: Cross-generator and robustness experiments.

Week 4: Analysis and error review.

Week 5: Final report and presentation.